

Henry Honglei Zheng

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Education

Stanford University

Physics Ph.D, Advisors: Leonardo Senatore, Eva Silverstein
Thesis title: Cosmology with the Effective Theory of Large Scale Structures

Stanford, CA
Sep 2019 – Present

University of Chicago

B.A. Physics with Honors

Chicago, IL
Sep 2015 – Jun 2019

Current Research Focus

Cosmology In broad terms my research in cosmology focuses on modeling the evolution of the universe with the Effective Field Theory of Large Scale Structures (EFTofLSS). I have developed efficient numerical packages for evaluating correlation functions of galaxies computed with EFTofLSS. Currently I am working on progressing Bayesian inference methods and sampling algorithms to fit models to latest galaxy survey data.

Machine Learning At the intersection of cosmology and AI, we can train neural networks to learn the computation of correlation functions, leveraging fast inference time compute. I am also working on new dictionary learning methods for mechanistic interpretability using variational autoencoders to disentangle correlation structures among features. In addition, I am seeking to understand how Chain-of-Thought (CoT) reasoning capabilities manifests e.g. through Group Relative Policy Optimization (GRPO) using the latent embedding space learned by sparse autoencoders. Lastly I am benchmarking a new response + evaluation feedback prompting loop with state-of-the-art reasoning models to improve model responses to research level theoretical physics problems.

Research Experience

Supervised Program for Alignment Research 2025

Feb 2025 – Present

- Under mentor Yuxiao Li: Developing a hybrid variational autoencoder and sparse autoencoder framework for small language models with goal of maintaining high interpretability and scalability.

Researcher, SITP and KIPAC

Sep 2019 – Present

- Developing and implementing the theory of bias and prior volume effect correction in cosmological Bayesian analysis resulting from noisy data and model misspecification (Paper in progress).
- Developing an emulator of PyBird, a python code for computing one-loop effective field theory predictions of two and three point correlators with application to various JAX based samplers (Paper in progress).
- Constructing a new method of solving partial differential equations using large language models with scalable inference time compute.
- Reduced uncertainty from weight initialization of small multilayer perceptron (MLP) models used to reweight phase space distributions in Large Hadron Collider (LHC) jet physics data using a new optimization algorithm [5].
- Benchmark of future cosmological surveys' neutrino sensitivity by comparing neutrino mass bounds obtained from fisher forecasts under various model assumptions [6].
- Forecasting next generation galaxy surveys Dark Energy Spectroscopic Instrument (DESI) and MegaMapper's cosmological parameter constraints with Fisher methodology. Constructed theoretically motivated priors that improve constraints on hard-to-measure non-Gaussianities by a factor of 2 [3].

- Developed an algorithm for solving generic one-loop quantum field theory integrals with complex masses in 3 dimensions. Optimized in both Cython and in Rust to achieve $5000\times$ speedup compared to previous state-of-the-art when applied to the one-loop bispectrum in the EFTofLSS [1].
- Built python and mathematica pipeline for Fisher analysis to forecast cosmological parameter constraints using one-loop galaxy powerspectrum and bispectrum for future surveys such as DESI and Megamapper [3].
- Performed Monte Carlo Markov Chain (MCMC) analysis on Stanford cluster Sherlock using a suite of data sets to constrain Rock N' Roll and Early Dark Energy models of the early universe [4].

Undergraduate Researcher, University of Chicago

Jun 2017 – Jun 2019

- Worked on high voltage electronics design, serial communication design, physics data analysis in ROOT and detector simulations using GEANT4 [2].

SLAC Summer Intern

May – Sep 2017

- Modeled ion diffusion through expitaxial thin films using time series X-ray reflectivity data to measure the ion diffusion coefficient [7].

Talks and Teaching

- Galaxy meets Quantum Chromodynamics (QCD) workshop: Computing the one-loop bispectrum in redshift space of galaxies
- ETH QCD Group Meeting Informal Talk: Brief Introduction to Modern Cosmology
- ETH Cosmology Group Meeting Informal Talk: Loop Integrals in EFTofLSS
- Teaching Assistant for Partial Differential Equations of Mathematical Physics, The Origin and Development of the Cosmos, Advanted Topics in Quantum Mechanics, Qunqum Field Theory II, Back of the Envelope Physics

Publications

- [1] "Efficiently evaluating loop integrals in the EFTofLSS using QFT integrals with massive propagators". In: *JHEP* 01 (2024), p. 002. DOI: [10.1007/JHEP01\(2024\)002](https://doi.org/10.1007/JHEP01(2024)002). arXiv: [2212.07421](https://arxiv.org/abs/2212.07421) [[astro-ph.CO](#)].
- [2] "Search for millicharged particles in proton-proton collisions at $s=13$ -TeV". In: *Physical Review D* 102.3 (Aug. 2020). ISSN: 2470-0029. DOI: [10.1103/physrevd.102.032002](https://doi.org/10.1103/physrevd.102.032002). URL: <http://dx.doi.org/10.1103/PhysRevD.102.032002>.
- [3] "Peeking into the next decade in Large-Scale Structure Cosmology with its Effective Field Theory". In: (July 2023). arXiv: [2307.04992](https://arxiv.org/abs/2307.04992) [[astro-ph.CO](#)].
- [4] "The Hubble Tension in Light of the Full-Shape Analysis of Large-Scale Structure Data". In: *JCAP* 05 (2021), p. 072. DOI: [10.1088/1475-7516/2021/05/072](https://doi.org/10.1088/1475-7516/2021/05/072). arXiv: [2006.12420](https://arxiv.org/abs/2006.12420) [[astro-ph.CO](#)].
- [5] "Optimizers for Stabilizing Likelihood-free Inference". In: (Jan. 2025). arXiv: [2501.18419](https://arxiv.org/abs/2501.18419) [[hep-ph](#)].
- [6] "Neutrino masses from large-scale structures: Future sensitivity and theory dependence". In: *Phys. Dark Univ.* 47 (2025), p. 101803. DOI: [10.1016/j.dark.2024.101803](https://doi.org/10.1016/j.dark.2024.101803). arXiv: [2412.04959](https://arxiv.org/abs/2412.04959) [[astro-ph.CO](#)].
- [7] "Ultrathin Layered SnSe Nanoplates for Low Voltage, High-Rate, and Long-Life Alkali-Ion Batteries". In: *Small* 13.46 (2017).

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